

Motion Synergy API 3rd-party Software Libraries

Table of Contents

Motion Synergy API 3rd-party Software Libraries	1
1 Revision History.....	2
2 Introduction	3
2.1 Purpose	3
2.2 Scope	3
2.3 Audience.....	3
3 3rd-party Software Libraries	4
3.1 Boost C++ Library	4
3.2 C++ Standard Library (Windows)	5
3.3 Lua Scripting Library.....	6
3.4 NumericInput Control Library	7
3.5 OxyPlot - Plotting Library for .NET	8
3.6 ZLib Library	9
4 License Text.....	11
4.1 Motion Synergy API License Text	11
4.1.1 <i>Boost C++ Library License Text</i>	<i>11</i>
4.1.2 <i>C++ Standard Library (Windows) License Text</i>	<i>11</i>
4.1.3 <i>Lua Scripting Library License Text</i>	<i>16</i>
4.1.4 <i>NumericInput Control Library License Text.....</i>	<i>16</i>
4.1.5 <i>OxyPlot - Plotting Library for .NET License Text.....</i>	<i>16</i>
4.1.6 <i>ZLib Library License Text.....</i>	<i>17</i>
4.2 Standard License Text	17
4.2.1 <i>Standard MIT License.....</i>	<i>17</i>

1 Revision History

Issue	Date	By	Description
A	21 OCT 2021	Mark Gladding	Initial revision.
B	03 NOV 2021	Mark Gladding	Added NumericInput Control Library.

2 Introduction

2.1 Purpose

This document details the 3rd-party Software Libraries and their associated licenses used by the MotionSynergyGUI and MotionSynergyAPI dll.

2.2 Scope

This document covers the following information related to 3rd-party libraries:

- Identification of 3rd-party libraries.
- Defines the functional and performance requirements for each 3rd-party library (covered by the *Intended Use* and *Selection Justification* sections for each library).
- Selection and validation strategy for each 3rd-party library.
- Details the license(s) for each 3rd-party library.
- Documents the title, version, manufacturer for each 3rd-party library.
- Assigns a unique designator for each 3rd-party library.
- Software (dependencies) necessary to support the proper operation of the 3rd-party library.

2.3 Audience

This document provides a reference of selected 3rd-party libraries and a record of the selection criteria used for each 3rd-party library.

The intended audience for this document includes:

- Software team members charged with the design and implementation of the software.
- Software and system testers charged with verifying the correct operation of the software.
- Future software developers charged with maintenance of the software.

3 3rd-party Software Libraries

3.1 Boost C++ Library

ID

IMP-SOUP-BOOST

Title

Boost C++ Library

Version

1.66.0

Manufacturer

Not Applicable (Open Source)

Description

Boost is a set of high-quality, free, peer-reviewed, open-source cross-platform C++ libraries. It works well with the C++ Standard Library. Use of Boost reduces initial development costs, results in fewer bugs, reduces reinvention-of-the-wheel, and cuts long-term maintenance costs.

URL<https://www.boost.org/>**Intended Use**

Boost has two primary purposes:

1. Provide an abstraction layer which hides operating system specifics. This allows cross-platform (e.g. Linux and Windows) applications to be developed with minimal effort.
2. Provides a set of general purpose libraries which reduce the amount of code needed to develop a specific application. Some commonly used libraries include asio (network and serial communications, timers), regex (regular expressions), property trees (configuration file support), etc.

Selection Justification

Following is the list of attributes used in the selection of this library:

- It is a mature library with wide adoption across the software industry.
- It has active development and patches.
- Royalty-free license conditions permitting use in a closed source, commercial product.
- Cross-platform support for both Linux and Windows.
- Minimal or no dependencies on other libraries.
- Functions correctly within the CPU, memory and persistent storage resource constraints of the Central Controller software runtime environment.

License[Boost Software License, Version 1.0](#)**Dependent Units****Dependencies**

- IMP-SOUP-CPPLIBLINUX or IMP-SOUP-CPPLIBWIN (depending on the target platform)

Limitations

No known issues impacting Dependent Units.

Validation Strategy

Validated as part of the unit/integration tests of dependent Software Units in the required context.

3.2 C++ Standard Library (Windows)

ID

IMP-SOUP-CPPLIBWIN

Title

C++ Standard Library (Windows)

Version

Microsoft Visual Studio 2019 v16.11.6

Microsoft Visual C++ Toolset v142

Namespace

std

Manufacturer

Microsoft

Description

Provides an implementation of the standard libraries associated with the C++ language. These libraries include:

- C++ Standard Library
- C Standard Library
- C Maths Library

URL

<https://visualstudio.microsoft.com/>

Intended Use

The C++ standard library has two primary purposes:

1. Provide an abstraction layer which hides operating system specifics. This allows cross-platform (e.g. Linux and Windows) applications to be developed with minimal effort.
2. Provides a set of general purpose libraries which reduce the amount of code needed to develop a specific application. Some commonly used libraries include containers, shared pointers, threading, etc.

Selection Justification

Included as part of the Microsoft Visual Studio C++ toolset, a high quality, continuously maintained product used industry-wide for development on the Microsoft Windows platform.

License

[Microsoft Visual Studio License](#)

Dependent Units**Dependencies**

None.

Limitations

No known issues impacting Dependent Units.

Validation Strategy

Validated as part of the unit/integration tests of dependent Software Units in the required context.

3.3 Lua Scripting Library

ID

IMP-SOUP-LUA

Title

Lua Scripting Library

Version

5.3.4

Manufacturer

PUC-Rio

Description

Lua is a powerful, efficient, lightweight, embeddable scripting language. Lua is dynamically typed, runs by interpreting bytecode with a register-based virtual machine, and has automatic memory management with incremental garbage collection, making it ideal for configuration, scripting, and rapid prototyping. Several benchmarks show Lua as the fastest language in the realm of interpreted scripting languages. Lua is fast not only in fine-tuned benchmark programs, but in real life too.

URL

<https://www.lua.org/>

Intended Use

Lua provides a scripting language for creating workflows to coordinate the control of multiple hardware devices. Typical workflows include initialising hardware, performing a power on self test, running an assay, performing a maintenance function such as emptying waste, etc. Lua scripting is not intended to perform hard realtime operations (i.e. requiring sub-50ms timing precision) which involve coordination of two or more hardware devices due to latency and jitter between the script command execution and the hardware control.

Selection Justification

Lua has been used in many industrial applications (e.g., Adobe's Photoshop Lightroom), with an emphasis on embedded systems (e.g., the Ginga middleware for digital TV in Brazil) and games (e.g., World of Warcraft and Angry Birds). Lua is currently the leading scripting language in games. Lua has a solid reference manual and there are several books about it. Several versions of Lua have been released and used in real applications since its creation in 1993.

Lua was selected as it can be run on both the Central Controller and the resource-constrained embedded environment of the Subsystem Controller. Currently Lua is only used on the Central Controller.

License

[MIT License](#)

Dependent Units**Dependencies**

- IMP-SOUP-CPPLIBLINUX or IMP-SOUP-CPPLIBWIN (depending on the target platform)

Limitations

No known issues impacting Dependent Units.

Validation Strategy

Validated as part of the unit/integration tests of dependent Software Units in the required context.

3.4 NumericInput Control Library

ID

IMP-SOUP-NUMERIC-INPUT

Title

NumericInput Control Library

Version

0.5.5

Namespace

Gu.Wpf.NumericInput

Manufacturer

Johan Larsson

Description

WPF TextBox for numeric input. Includes controls DoubleBox, IntBox, DecimalBox, FloatBox and ShortBox.

URL

<https://github.com/GuOrg/Gu.Wpf.NumericInput>

Intended Use

Provides controls for numeric input via the MotionSynergyGUI.

Selection Justification

Following is the list of attributes used in the selection of this library:

- It is a mature library with wide adoption across the software industry.
- It has active development and patches.
- Royalty-free license conditions permitting use in a closed source, commercial product.
- Cross-platform support for Linux, QNX and Windows.

- Minimal or no dependencies on other libraries.
- Functions correctly within the CPU, memory and persistent storage resource constraints of the MotionSynergyGUI software runtime environment.

License[MIT License](#)**Dependent Units**

- MotionSynergyGUI

Dependencies

None

Limitations

No known issues impacting Dependent Units.

Validation Strategy

Validated as part of the unit/integration tests of dependent Software Units in the required context.

3.5 OxyPlot - Plotting Library for .NET

ID

IMP-SOUP-OXYPLOT

Title

OxyPlot - Plotting Library for .NET

Version

2.1.2

Namespace

OxyPlot.Core, OxyPlot.Wpf, OxyPlot.Wpf.Shared

Manufacturer

OxyPlot contributors

Description

OxyPlot is a cross-platform plotting library for .NET. A number of support packages are available to support a specific UI framework.

URL<https://oxyplot.github.io/>**Intended Use**

Provides plotting capabilities in the MotionSynergyGUI.

Selection Justification

Following is the list of attributes used in the selection of this library:

- It is a mature library with wide adoption across the software industry.
- It has active development and patches.
- Royalty-free license conditions permitting use in a closed source, commercial product.

- Cross-platform support for Linux and Windows.
- Minimal or no dependencies on other libraries.
- Functions correctly within the CPU, memory and persistent storage resource constraints of the MotionSynergyGUI software runtime environment.

License[MIT License](#)**Dependent Units**

- MotionSynergyGUI

Dependencies OxyPlot.Core package implements the core plotting related APIs. However, to use OxyPlot with a UI framework, support packages for the UI framework are required. For example, OxyPlot.Wpf and OxyPlot.Wpf.Shared packages are required for OxyPlot to work with WPF UI framework.

Limitations

No known issues impacting Dependent Units.

Validation Strategy

Validated as part of the unit/integration tests of dependent Software Units in the required context.

3.6 ZLib Library

ID

IMP-SOUP-ZLIB

Title

ZLib Library

Version

1.2.11

Manufacturer

Jean-loup Gailly and Mark Adler

Description

zlib is designed to be a free, general-purpose, legally unencumbered -- that is, not covered by any patents -- lossless data-compression library for use on virtually any computer hardware and operating system. The zlib data format is itself portable across platforms.

URL

<https://www.zlib.net/>

Intended Use

zlib provides the compression algorithms used when creating zip archives of log files.

Selection Justification

Following is the list of attributes used in the selection of this library:

- It is a mature library with wide adoption across the software industry.

- It has active development and patches.
- Royalty-free license conditions permitting use in a closed source, commercial product.
- Cross-platform support for both Linux and Windows.
- Minimal or no dependencies on other libraries.
- Functions correctly within the CPU, memory and persistent storage resource constraints of the Central Controller software runtime environment.

License

[ZLib license](#)

Dependent Units

Dependencies

None.

Limitations

No known issues impacting Dependent Units.

Validation Strategy

Validated as part of the unit/integration tests of dependent Software Units in the required context.

4 License Text

4.1 Motion Synergy API License Text

4.1.1 Boost C++ Library License Text

Boost Software License - Version 1.0 - August 17th, 2003

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4.1.2 C++ Standard Library (Windows) License Text

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zlib.h -- interface of the 'zlib' general purpose compression library
version 1.2.11, January 15th, 2017

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